Lesson notes · for you

Pack the truck

Optimization · Level 1 · Lesson 1

Time
20 min

Materials
Paper and a pen; Six small things from your bag, or six paper cards

Learning goals

- Say the rule and the goal of a packing puzzle.
- Check whether a load keeps the rule.
- Find the best load, and think about what stays behind.

Activity: Activity: Pack the bag

1. Cards (5 min): Write six things you might carry, such as a book, a water bottle, lunch, a jumper, a ball and a pencil case. Give each a weight from 1 to 5 and points from 1 to 10.
2. Pack (8 min): Put cards together and add up their weights. Any load that adds up to more than 10 doesn't count. Write down three loads that fit.
3. Best (5 min): Add up the points of each load that fits. Circle the one with the most points.
4. Share (2 min): Show your partner your best load. Can they find one with more points that still fits?

No cards? Tear a page into six pieces. No device needed.

Discussion

- Why didn't the quick way give the best load?
- Why doesn't a load that is too heavy count?
- Who should decide what stays behind?

Check yourself

- The best truck load is ___ and ___, with ___ points.

Try it this week
Pack your school bag like a truck. Give each thing a weight (1 to 5) and points (how much you need it), and choose a limit, like 10. Find the load with the most points that stays under the limit, then ask a friend or someone at home to beat it.

Worksheet

Pack the truck

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Name _____ Class _____ Date _____

1 **You load the desks (500 kg, 33 points) and the water filter (300 kg, 15 points): 800 kg and 48 points. Is it the best load?**

◯ Yes: it keeps the rule ◯ No: desks and books give more points ◯ No: it's too heavy

2 **A load that is too heavy can still be the best load if it has the most points.**

◯ True ◯ False

3 **Match each word to what it means in the truck puzzle.**

The rule	• • • •	The six loads
The goal	• • • •	The most points
The choices	• • • •	No more than 1,000 kg

4 **Which loads fit on the truck together (1,000 kg or less)? Choose all that fit.**

☐ Solar panels and desks (600 + 500) ☐ Desks and books (500 + 500)

☐ Solar panels and the water filter (600 + 300) ☐ Books, water filter and footballs (500 + 300 + 200)

☐ Solar panels, books and chalk (600 + 500 + 100)

5 **Put the loads in order, from the most points to the fewest.**

☐ Chalk and paper (3)

☐ Footballs (8)

☐ Water filter (15)

☐ Books (32)

☐ Desks (33)

☐ Solar panels (42)

✎ **Think of a time you couldn't take everything: a bag, a trip, pocket money. What was your rule, what was your goal, and what stayed behind?**